① Programmed I/O :

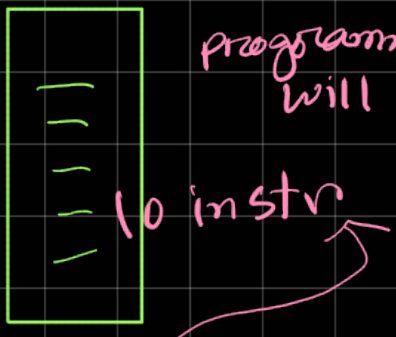
② Interrupt Driven I/O

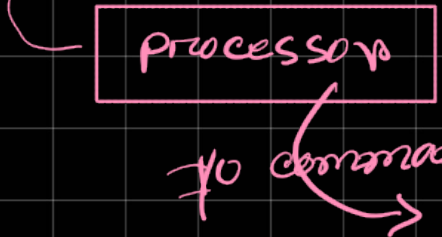
③ DMA

IO instr → 1st

IO command

↓

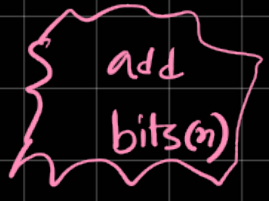




Address Bus

Addressing Mode :

↓ of I/O device.



2^n address generate

32 bits Address but



memory mapped IO



alternative → same address bits used separately.



Isolated IO

IO mapped IO

Input/read

Output/write

Load

DMA → feedback
returned as interruption

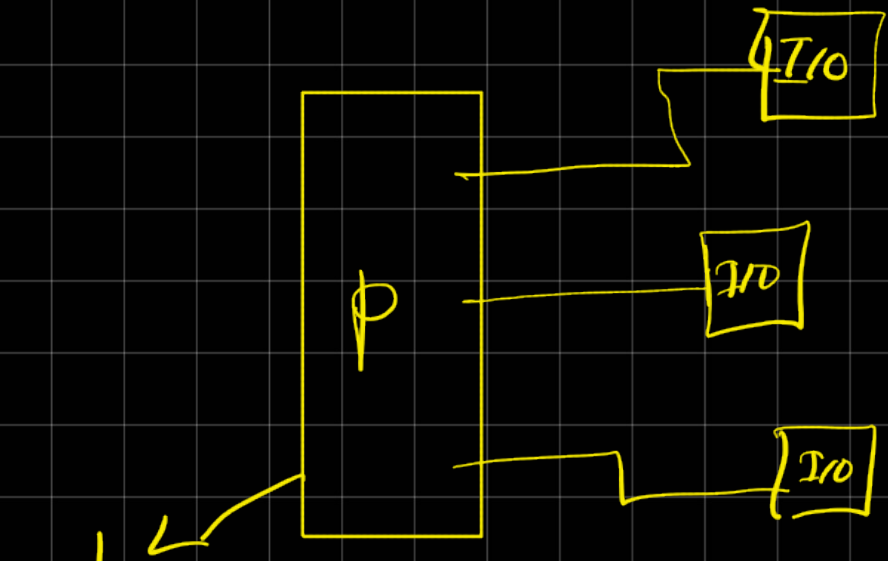
Response to an interrupt:

↳ problem → multiple devices
multiple ints.

How would processor understand which one → identification problm, priority.

1 soln:

Mutiple interrupt lines.



but valuable & limited.

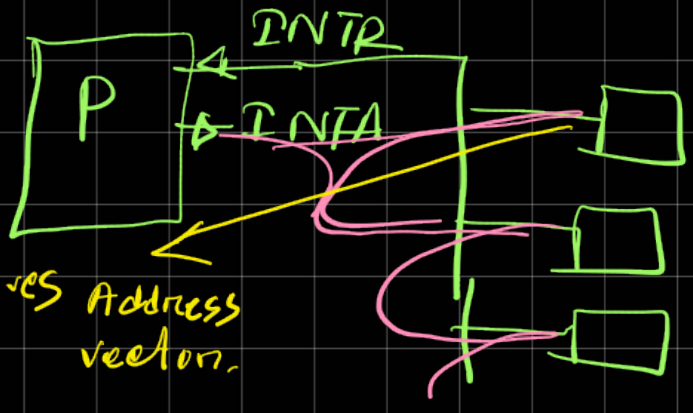
② Software poll. → IH funct
↳ Test — Add

IH → generic
↳ 1st execute

A — Acknowledge (if intercepted)

③ Hardware poll

Daisy chain interruption pin.



if found gives Address vector.

Encoder :